

Arnav Patil

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EDUCATION

University of Toronto

Sept 2023 – Apr 2028

BASc. in Computer Engineering with PEY Co-op

Toronto, ON

- **GPA: 3.63/4.00** with recognition on Dean's Honours List
- **Minor in Engineering Business**

EXPERIENCE

Digital Design Engineer Intern

May 2026 – Present

Marvell Technology

Toronto, ON

- Writing Verilog RTL for PHY-layer SerDes die-to-die interconnect IP targeting TSMC N2 and N3 process nodes
- Debugging simulation infrastructure and testbenches in VCS to support functional validation & silicon bring-up
- Developing AI-assisted design-review workflows using ChipAgents to automate specification analysis and review

Teaching Assistant

Sept 2025 – Dec 2025

University of Toronto

Toronto, ON

- Led tutorials for **APS100 Orientation to Engineering**, reinforcing engineering concepts through discussions
- Mentored first-year engineering students on academic planning, study strategies, and professional development

FPGA Engineer Intern

May 2025 – Aug 2025

Intelligent Sensory Microsystems Lab

Toronto, ON

- Developed FPGA-based modules for novel CMOS time-of-flight sensors in Prof. Roman Genov's **ISML** group
- Achieved 800 Mb/s sensor readout throughput by designing high-speed RTL pipelines on Xilinx Artix-7 FPGAs
- Implemented SPI and I2C controller IP in Verilog to enable sensor initialization, configuration, and control

PROJECTS

MOSAIC: Multi-Objective Smooth AnalytICal Placer |

- Macro placement for the **2026 Macro Placement Challenge** co-hosted by Partel and Hudson River Trading
- Achieved a 10.3% improvement on RePlAcE on IBM ICCAD04 benchmarks, averaging a 99 s/bench runtime
- Implemented differentiable approximations of TILOS MacroPlacement functions for gradient-based optimization

SpecNet: Deep Learning for Analog Circuit Design |

- Developed an ML pipeline that predicts transistor sizing parameters from target analog circuit specifications
- Achieved 98% accuracy within 5% of optimal dimensions, greatly reducing number of SPICE simulations required
- Automated data generation by parsing LTspice simulation outputs into structured datasets for supervised learning

DE1-SoC Arcade Game |

- Implemented a blackjack game in Verilog on an FPGA, managing FSMs for game logic and player interaction
- Simplified hardware peripheral control from top-level modules by writing API-like wrapper RTL over Altera IP
- Fixed version compatibility issues in provided IP by writing a Python script to re-format memory files (.mif)

Nios-V Ultrasound System |

- Integrated an ultrasonic sensor and servo motor into the FPGA-based Nios-V soft processor via GPIO ports
- Designed & implemented BJT pull-up/pull-down networks for safe voltage conversion between FPGA and sensor
- Implemented timer-based polling in C to calculate object distance from 20 cm to 2 m within 2 cm precision

TECHNICAL SKILLS

Languages: Verilog, C, C++, Python, csh

Tools: Intel Quartus Prime, AMD Vivado, Cadence Virtuoso, Synopsys VCS

Hardware: Computer Architecture, Digital & Analog Electronics, Computer Hardware

Software: Operating Systems, Embedded Systems, Machine Learning